

Recognising Geometric Series, p-Series, and the nth-Term Test

Answer Key

AP Calculus BC

Quick Answers

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| 1. the sum = $\frac{3}{2}$, the written observation = — | 5. the sum = $\frac{\pi^4}{90}$, the written explanation = — |
| 2. the sum = $\frac{\pi^2}{6}$ | 6. the limit of the terms = 1, the written conclusion = — |
| 3. the limit of the terms = $\frac{3}{5}$, the written conclusion = — | 7. the fraction = $\frac{5}{11}$, the written argument = — |
| 4. the distance after the first drop = 6, the total distance = 8.0 m | 8. the sum of the first series = 3, the limit for the third series = 1, the written classification = — |
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What is verified

10 of 16 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 1, 3, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus BC

LIM-7 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 16 tagged checks.

1. Partial sums of $\sum_{n=0}^{\infty} (1/3)^n$ are 1, 1.3333, 1.4444, 1.4815. Find the exact sum and say what the table suggests.

Solution: The ratio is $r = \frac{1}{3}$ and the first term is 1, so the series converges and

$$S = \frac{\text{first term}}{1 - r} = \frac{1}{1 - \frac{1}{3}} = \frac{1}{\frac{2}{3}}.$$

So the sum is

$\frac{3}{2}$

Written half — instructor-judged. The table shows the partial sums climbing and flattening toward about 1.5, which is evidence for the limit but not a proof of it: no finite list of partial sums can rule out a later change of behaviour. The geometric-series formula is what proves the value. Full credit makes that distinction.

2. Find the exact sum of $\sum_{n=1}^{\infty} 1/n^2$.

Solution: This is the p -series with $p = 2 > 1$, so it converges, and its sum is the standard value

$$\frac{\pi^2}{6}$$

3. $\sum \frac{3n^2+1}{5n^2+2n}$. (a) Find the limit of the terms. (b) What does the nth-term test conclude?

Solution (a): Divide numerator and denominator by n^2 : the terms become $\frac{3 + 1/n^2}{5 + 2/n}$,

which tends to

$$\lim_{n \rightarrow \infty} a_n = \frac{3}{5}$$

Solution (b) — instructor-judged. The nth-term test says: if $\lim a_n \neq 0$ then $\sum a_n$ diverges. Here the limit is $\frac{3}{5}$, so the series diverges. Had the limit been 0, the test would have concluded nothing at all — it never proves convergence, and $\sum \frac{1}{n}$ is the standing counterexample. Full credit states the test in that one direction and gives both halves.

4. A ball dropped from 2 m rebounds to $\frac{3}{5}$ of its previous height forever. Total vertical distance?

Solution: The first drop is 2 metres. After that, the n -th bounce contributes an up-and-down pair of $2 \cdot 2 \left(\frac{3}{5}\right)^n = 4 \left(\frac{3}{5}\right)^n$ metres, so the bouncing accounts for

$$\sum_{n=1}^{\infty} 4 \left(\frac{3}{5}\right)^n = \frac{4 \cdot \frac{3}{5}}{1 - \frac{3}{5}}.$$

That bouncing distance is

$$6 \text{ m}$$

Adding the initial drop gives a total of

$$8 \text{ m}$$

5. Partial sums of $\sum 1/n^4$ are 1, 1.0625, 1.0748, 1.0788. (a) Exact sum. (b) Why does it converge?

Solution (a): This is the p -series with $p = 4$, whose sum is the standard value

$$\frac{\pi^4}{90}$$

Solution (b) — instructor-judged. A p -series $\sum \frac{1}{n^p}$ converges precisely when $p > 1$, and here $p = 4$. The table is that rule made visible: the terms shrink so fast that four of them already put the partial sum within 0.005 of the limit, whereas the $p = 1$ series creeps upward forever. Full credit ties the exponent to what the table is doing.

6. $\sum n \sin(1/n)$. (a) Find the limit of the terms. (b) State the conclusion and explain.

Solution (a): Substituting $u = 1/n$ turns the terms into $\frac{\sin u}{u}$ with $u \rightarrow 0$, a limit

already known to be

$$\lim_{n \rightarrow \infty} n \sin\left(\frac{1}{n}\right) = 1$$

Solution (b) — instructor-judged. The terms approach 1, not 0, so by the nth-term test the series diverges. The reason they fail to shrink is that the two factors cancel: $\sin(1/n)$ shrinks like $1/n$, and the factor n grows at exactly that rate, so the product settles at 1. Full credit gives that cancellation, not just the limit.

7. $0.\overline{45} = \sum_{n=1}^{\infty} 45/100^n$. (a) Write it as a fraction. (b) Why is every repeating decimal rational?

Solution (a): First term $\frac{45}{100}$, ratio $\frac{1}{100}$:

$$S = \frac{45/100}{1 - 1/100} = \frac{45}{99}.$$

In lowest terms that is

$$\boxed{\frac{5}{11}}$$

Solution (b) — instructor-judged. Any repeating decimal has a block of k digits repeating, so it equals $\sum_{n \geq 1} \frac{B}{10^{kn}}$ where B is the block read as an integer. That is a geometric series with rational first term and rational ratio $10^{-k} < 1$, so it converges and its sum $\frac{B}{10^k - 1}$ is a ratio of integers. Full credit gives the general argument, not a second example.

8. (i) $\sum (2/3)^n$, (ii) $\sum 1/\sqrt{n}$, (iii) $\sum \frac{n}{n+4}$. (a) Sum (i). (b) Limit for (iii). (c) Classify all three.

Solution (a): Geometric with first term 1 and $r = \frac{2}{3}$, so the sum is $\frac{1}{1 - 2/3}$.

$$\boxed{\sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n = 3}$$

Solution (b): Dividing through by n gives $\frac{1}{1 + 4/n}$, so the limit is

$$\boxed{\lim_{n \rightarrow \infty} \frac{n}{n+4} = 1}$$

Solution (c) — instructor-judged. (i) converges: geometric with ratio $\frac{2}{3}$, and $|r| < 1$. (ii) diverges: it is the p -series with $p = \frac{1}{2} \leq 1$. (iii) diverges: its terms approach 1, not 0, so the nth-term test applies. Full credit names the deciding quantity in each case — the ratio, the exponent, the limit of the terms — rather than only the verdict.